



What Is a Bug?

Quick facilitation notes & answer key

Goal

Students learn the debugging method — SHOULD, DOES, FIX — and apply it to find and fix a wrong repeat count in a loop (C3.2). The challenge separates total count from repeat count when two blocks are inside.

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

A block-coding app (Scratch-style) to show a buggy loop and fix it live; the printed worksheet + a pencil per student.

Run it (about 30 min)

1. Define bug + debugging; introduce SHOULD → DOES → FIX with the jump example.
2. Run repeat 3 (3 jumps) vs the goal (5); fix to repeat 5 together.
3. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Share answers; land the big idea: name the goal first, then find the wrong part.

Answer key (checked by the run-gate)

Ex 1 — repeat 6 (repeat 4 is too few).

Ex 2 — too many; should be repeat 4.

Ex 3 — repeat 8.

Ex 4 (challenge) — repeat 3 ($2 \text{ claps} \times 3 = 6$); repeat 6 gives 12.

Watch for & differentiate

Common slip: not checking the goal first, so any number 'looks' fine.

Support: write the GOAL number, then compare to the repeat number.

Extend: ask the repeat for 10 claps with 2 inside (5).

Success looks like

The student states the goal, compares it to what the code does, and fixes the repeat number — including when two blocks inside change the total (Ex 4).